ON THE PARADOXICAL TIME-STRUCTURES OF GÖDEL*

HOWARD STEIN

Case Western Reserve University

Gödel's conclusion that time-travel is possible in his models of Einstein's gravitational theory has been questioned by Chandrasekhar and Wright, and treated as doubtful in the recent philosophical literature. The present note is intended to remove this doubt: a review of Gödel's construction shows that his arguments are entirely correct; and the objection is seen to rest upon a misunderstanding. Computational points treated succinctly by Gödel are here presented in fuller detail. The philosophical significance of Gödel's results is briefly considered, and a set of technical questions is posed whose answers would clarify this significance.

Gödel ([4], [5]) discovered, in 1949, the existence of models for Einstein's gravitational theory having a remarkable temporal structure; their most striking characteristic is that they allow the possibility of time-travel. Gödel's conclusions have been questioned by Chandrasekhar and Wright [1], and have accordingly been referred to in the recent philosophical literature (cf. [6], pp. 201–202) as of uncertain validity. The present note is intended to remove this doubt: careful review of Gödel's construction shows that his arguments are entirely correct; and the objection in [1] is easily seen to rest upon a misunderstanding. The technical review of Gödel's construction is given in an Appendix (sections **IV–VII**). In the course of the review, some computational points that Gödel treats very succinctly are presented in fuller detail; it is hoped that this will make Gödel's construction more accessible to philosophers concerned with such matters. The principal results are summarized in section **I**; in sections **II** and **III**, the philosophical significance of Gödel's results is briefly considered, and a set of technical questions posed whose answers would greatly clarify this significance.

I

The space-time of Gödel's model is Euclidean in its topology; its Minkowski metric, in terms of a particular global system of coordinates z_0, z_1, z_2, z_3, is specified by equation (1) (section **IV** below). The world-lines determined by constant values of z_1, z_2, z_3, are time-like geodesics, parametrized by z_0 as proper time; we shall call them "z_0-lines." If the cosmic masses—i.e. those which appreciably influence the gravitational field (and hence the space-time metric)—are taken to be distributed uniformly through space-time, and if the world-lines of those masses are taken to be the z_0-lines, then Einstein's gravitational equations are satisfied; in this sense, Gödel's space-time metric is, for general relativity, a possible structure of the physical universe.

To obtain the salient results concerning this space-time structure, Gödel introduces a transformation to a kind of "cylindrical coordinates." More precisely, if we consider the four-dimensional manifold ("hypercylinder") obtained as the Cartesian

* Received July, 1969. Work supported by the National Science Foundation.

This content downloaded from 147.026.011.080 on July 17, 2017 06:50:22 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

product of three lines and a circle, with coordinates ρ, τ, η, on the lines, and with "angular coordinate" (in radian measure) 2θ on the circle (so that θ varies from 0 to π in one circuit), then equations (2), (3), (4), (5), with side-condition (4′) (section **IV** below), define a continuous and differentiable mapping f of this hypercylinder into space-time. The properties of the mapping f allow us to regard ρ, θ, τ, η, as "cylindrical coordinates" on space-time: if we impose the restriction $\rho \geq 0$, then all these "coordinates" except θ are well-defined everywhere on space-time; and for $\rho > 0$, θ too is well-defined and the four together constitute a system of differentiable coordinate-functions. (The pair ρ and θ—which depend only upon z_1 and z_2—are related essentially like the radius and angle in a polar coordinate system: the locus $\rho = 0$, on which θ is undetermined, is the locus $z_1 = z_2 = 0$—the origin of the z_1z_2-plane.) The "τ-lines" of this coordinate system—i.e. the lines on which ρ, θ, η, are constant—coincide with the z_0-lines; and on any such line τ, like z_0, measures proper time.

Everything so far looks quite tame: we have two coordinate systems, each of which seems to have a "time-coordinate" and three "space-coordinates."[1] But now we consider the "circles" of our system of cylindrical coordinates—the "θ-lines," defined by constant values of ρ, τ, η. These are, of course, closed curves, traversed once as θ goes from 0 to π. But when we express the space-time metric in terms of ρ, θ, τ, η (for the computations see section **V** below), we find that *provided the "radius" ρ is large enough, the tangent-vector to such a circle is time-like at each point.*

This is the crucial result upon which all Gödel's conclusions rest. It is easy to see that one of the two possible senses of description of a time-like world-line must be "future-oriented" at each point, and so must represent—on the usual understanding of the Einstein–Minkowski conceptual system—a "possible course-of-history" of a body. Suppose the body A has such a world-line as its history. From the point of view of an observer on one of the cosmic masses encountered by A—an observer whose world-line is a z_0-line—the body A is met only once in cosmic history. On the other hand, from the point of view of an observer on A, the encounter is periodically recurrent: indeed, all events are periodically recurrent, and the structure of time is "closed."—It is also easy to see that a very slight perturbation, which does not destroy the everywhere-time-like and future-pointing character of the world-line of A, can change that world-line from a circle to a helix; and a helix, moreover, that *winds about the z_0-lines in the backwards direction of their time.* Along such a "backwards-winding" line, time will not be "closed" and events will not recur. The experiences of an observer with such a world-line and those of his "cosmically normal" confrère will be related in the following remarkable way: During any single encounter, the two agree in their sense of the "passage" of time; they describe the sequence of events attendant upon the encounter in the same way.

[1] In fact, neither the manifolds $z_0 =$ constant nor the manifolds $\tau =$ constant are space-like: the metric of the former is indefinite (i.e. has a time-like dimension along with two space-like dimensions) at each point; that of the latter is indefinite beyond a certain radius. It is a simple consequence of the facts to be discussed that there exists *no* complete spatial cross-section of the Gödel space-time (as Gödel has noted in [4]).

This content downloaded from 147.026.011.080 on July 17, 2017 06:50:22 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

But from one encounter to the next, each finds that where he has grown older, his acquaintance has grown younger; and their recollections of past meetings are utterly different—indeed, if they exchange recollections systematically, each will discover the circumstances of the current meeting to be those described by the other, on their *previous* encounter, as what he then recalled of their *then previous* meeting. In short, over the intervals between visits, what is recollection for either is found to be correct prediction for the other.

It should not be supposed that the kinds of world-lines described exhaust the possibilities. It is, for example, very easy—from a purely geometrical point of view—to describe smooth world-lines, everywhere time-like and future-pointing, which for some parts of their course follow the cosmic flow—the z_0-lines—and for other parts have the characteristics of our "backwards-winding" lines. In particular, such piecing together can be done so as to make the backwards-winding parts of the line come into proximity with some of *its own* "normal" segments—thus making of the encounters we have considered *self*-encounters, and realizing the kind of bizarre history described long before by Weyl ([8], p. 249) and Reichenbach ([7], pp. 141–142), in which an observer comes to interact with his own past.[2]

[2] John Earman ([2], p. 221) questions the sense in which Gödel's universe admits the possibility of "time-travel." This sense seems to me clear from the propositions above. If "decisions" can be taken, and "free actions" performed, in such a universe—and if adequate energy-sources are available—then a person in such a world can decide, at any point in his career, to visit the space-time vicinity of any epoch of his own past; and (as closer inspection of the results stated will show) can make that visit with an arbitrarily small lapse of his own proper time. That this involves conceptual difficulties, as Earman suggests, must be freely admitted (what, for instance, if I assassinate my own great grandfather in his childhood?). The paradoxical character of the time-structure of his model is emphasized by Gödel in [5], where his chief aim *is to put in question* our ordinary notions about time.

It has been suggested to me that the notions of "travel into the past," "influencing the past," etc., call for a more critical analysis than I have here accorded them, taking into account e.g. the distinction between "open" and "closed" physical systems. I thoroughly agree that such discussion has great intrinsic interest; but I do not agree that the analysis in question is required for our present purposes; and I offer instead a metatheoretic reflection: The general problems of freedom *vs.* necessity, of determinism *vs.* indeterminism, and of the meaning of "cause," seem to me (like the problem of "the laws of nature"—see section **III** below) unlikely to receive a speedy conclusive resolution. But as, in methodological discussion (for example), we allow ourselves to speak of "testability" of hypotheses, or "measurability" of some physical quantity, without worrying that if the universe is deterministic we cannot "freely" institute the conditions of test or measurement, so in this species of cosmological context it appears to me permissible—and desirable—to adopt the following point of view, without initial regard to metaphysical considerations: Given a *macrocosmic* characterization of a universe—e.g. Gödel's—to be compared with the world of our experience, we treat it "in the small," or "locally," in terms of those common-sense notions that do in fact prove serviceable in our own quotidian "microcosmic" lives (in particular, in the present case, we assume the possibility "at will" of instituting technological procedures which result in any changes in our own state of motion that we *have no reason to believe excluded by physical principles*). We then explore the global possibilities within the postulated macrocosm, by systematic piecing together of such local possibilities.

This content downloaded from 147.026.011.080 on July 17, 2017 06:50:22 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

II

The propositions established quite justify Gödel's statement about this model ([4], p. 447):

> Every world line of matter occurring in the solution is an open line of infinite length, which never approaches any of its preceding points again; but there also exist closed time-like lines. In particular, if P, Q are any two points on a world line of matter, and P precedes Q on this line, there exists a time-like line connecting P and Q on which Q precedes P;[3] i.e., it is theoretically possible in these worlds to travel into the past, or otherwise influence the past.

What, then, of the objections of Chandrasekhar and Wright?

What these authors principally do in [1] is to integrate the differential equations for the geodesics of the metric (1): they succeed in giving all the coordinates z_i along an arbitrary time-like geodesic as elementary functions of proper time. They then raise (in effect) the question, whether any time-like geodesics lie on a three-dimensional space-time cylinder, the image under f of the set of those points of our hypercylinder for which the value of ρ is a specified constant R. Of course the z_0-lines are such geodesics; and so, more generally, are all the time-like "straight lines" in any z_0z_3-plane (these being totally geodesic and flat submanifolds of space-time). There is no *prima facie* reason to expect that there will be others; but Chandrasekhar and Wright show that they do in fact exist—for a certain bounded open interval of values of R. They also show that along any such geodesic, traversed in the future-pointing sense, the value of τ continually increases.

These are interesting results about the geometry of Gödel's model. But in using them to question Gödel's conclusions, Chandrasekhar and Wright incorrectly describe Gödel's argument as follows ([1], p. 345):

> By considering the special geodesics . . . which correspond to "circular orbits," r = constant, described uniformly, Gödel has drawn the conclusion that these orbits allow one "to travel into the past, or otherwise influence the past."

The condition r = constant—i.e., in our terms, ρ = constant—of course does not determine a "circle," but a three-dimensional subspace (the product of a circle and two copies of the real line). Gödel does in fact consider genuine "circles" lying in these subspaces—but does not claim that the circles are geodesics; and the conclusions that he draws from the geometry of these circles depend in no way upon their being geodesics. If the closed paths, or the backwards-winding paths, were geodesics, then one could "travel into the past" (after achieving a suitable initial state) simply by inertial motion (free fall); whereas Gödel in [5] speaks of a journey by rocket ship, and nowhere implies that the principal part of the journey is to be purely inertial.[4] Chandrasekhar and Wright, therefore, are mistaken when, having

[3] That this holds for any such P and Q can be seen as follows: (1) Our conclusions show that, for some Q, it holds for all such P near enough to Q. (2) Since extension of the backwards-winding path makes the encounters with a fixed z_0-line periodic, the assertion holds—for the Q of (1)—for all such P. (3) But Gödel's space-time is easily seen to be homogeneous ([4], p. 448); the same is therefore true of all Q.

[4] Earman (*loc. cit.*) makes this point, and expresses the opinion that the objection of Chandrasekhar and Wright rests on a misunderstanding. (According to Earman, "Gödel himself

This content downloaded from 147.026.011.080 on July 17, 2017 06:50:22 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

found that τ increases along their "circular" (more properly "cylindrical," or "helical") geodesics, they add ([1], p. 347): "This . . . fact seems to be contrary to some statements of Gödel from which he has drawn the conclusion we have quoted earlier."[5]

III

How strongly do Gödel's results bear upon our understanding of the temporal structure of the world, and upon the very notion of time? I think it is entirely plain that this bearing is strong indeed, *if* our universe has a structure akin to that of Gödel's model. There is no evidence that it does; although Gödel has argued ([5], p. 562) that there is no decisive evidence that it does not. Gödel also argues (*ibid.*) that "the mere compatibility" of such a structure "with the laws of nature" has consequences for the meaning of time in *any* world. This argument, however, rests upon (1) the assumption of a clear notion of "the laws of nature," and (2) the assumption that there are no such laws that rule out the Gödel universes. That the first of these assumptions is debatable—that the fundamental notion of a "law of nature" is problematic—is a commonplace of current philosophic discussion; and one may be pardoned for viewing pessimistically the short-range prospect of a clarification that would allow confident adjudication of Gödel's claim. But (perhaps paradoxically) the second assumption may admit a more decisive evaluation.

That the second assumption can be questioned is shown by the direct proposal that exclusion of the Gödel-type structures be taken as itself a "law of nature"—cf. Einstein, [3], p. 688:

> But does the assertion ["B is before A"] still make sense, if the points, which are connectable by the time-like line, are arbitrarily far separated from each other? Certainly not, if there exist point-series connectable by time-like lines in such a way that each point precedes temporally the preceding one, *and if the series is closed in itself.* . . .

---

states that the world lines which allow time travel are not geodesics." This seems to me not strictly correct: Gödel states (in [4]) that these are not "world lines of matter"; and (in [5]) he refers to travel by rocket ship. But there are geodesics other than the world-lines of matter; and rockets might be required only to initiate and to terminate a ballistic trajectory. However, Gödel certainly does *not* say that the world-lines that allow time-travel *are* geodesics; therefore I think Earman's comment is in substance quite right.)

[5] Gödel ([4], p. 449) expressly describes his "circles" as characterized by $\rho = R$, $\tau = 0$, $\eta = 0$; τ, therefore, certainly does not increase along them. Nevertheless, a question might arise as to whether these "circles" are closed curves. As θ varies from 0 to π, (4) allows us to trace the variation in z_0 required to keep $\tau = 0$ and $\rho = R$; and the possibility has to be considered that when this variation is made continuous, the final value of z_0 might not agree with its initial value. The side-condition (4′)—which Chandrasekhar and Wright do not in fact reproduce—determines z_0 univocally, and therefore assures that the values for $\theta = 0$ and $\theta = \pi$ do agree; but then one has to make sure that a continuous curve is described: i.e. that z_0 is not constrained by (4′) to jump, sometimes, from a value above τ to one below it, or vice versa. Were this to happen, the difference between τ and z_0 would have to attain the value π, excluded by (4′); so that, strictly speaking, (4′) could not then always be satisfied.—Now, this point is fully covered by the discussion in section V below. But Gödel passes over it in silence (presumably expecting the reader to verify that (4′) occasions no trouble). Perhaps Chandrasekhar and Wright, without turning direct attention upon the point, were somehow misled by it to doubt that Gödel's "circles" are closed.

This content downloaded from 147.026.011.080 on July 17, 2017 06:50:22 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

> Such cosmological solutions of the gravitation-equations ... have been found by Mr. Gödel. It will be interesting to weigh whether these are not to be excluded on physical grounds.

Gödel himself anticipates this proposal, and answers it as follows ([5], pp. 560–561):

> This state of affairs seems to imply an absurdity. [There follows a description of "time-travel" and "influencing of the past."] This and similar contradictions, however, in order to prove the impossibility of the worlds under consideration, presuppose the actual feasibility of the journey into one's own past. But the velocities which would be necessary in order to complete the voyage in a reasonable length of time are far beyond everything that can be expected ever to become a practical possibility. Therefore it cannot be excluded *a priori*, on the ground of the argument given, that the space-time structure of the real world is of the type described.[6]

The issue, so posed, is doubtless quite as hard to decide as that concerning the concept of "natural law" itself. What I should like to point out, however, is that there is a set of technical questions—possibly amenable to fruitful study—whose answers might contribute appreciable clarification. Gödel's results concern a cosmological model; in assessing the eligibility of that model, one is up against the well-known difficulty that only one cosmos is given us (so that what other ones are "physically possible" is moot). But the occurrence of closed, or backwards-winding, time-like world-lines is not *prima facie* dependent upon the large-scale characteristics of space-time. The questions that naturally present themselves are these: Consider either an arbitrary given cosmological model, or a model having a structure of one of the sorts ordinarily assumed to hold in the real world. Then: (1) is it ((a) ever, (b) always) possible to introduce into such a model a continuous deformation of the structure, leading through intermediate states, all compatible with Einstein's theory, to a state in which Gödel-type relationships occur? If the answer is yes, then: (2) can such a deformation of the structure be regarded as compatible with known laws of nature; i.e. can the deformation be effected by "forces" of a type at least not obviously excluded by known laws? Finally, and most strongly: (3) is it ("in principle") possible, by a *local* rearrangement of the distribution and flow of matter, to create the local possibility of time-travel? (Here "local" means within some bounded—though possibly large—region, within a cosmos whose structure is on the whole not Gödelian.)

Results affecting these questions might decisively clarify the significance of Gödel's results for the concept of time, independently of the problematic notion of law. This is clearest for the case that the answer to (3) proves affirmative. In this case, the suggestion that Gödelian structures be excluded *a priori* would have been shown—assuming the general theoretical framework not otherwise modified—to lead to inconsistency; in the very strongest version of affirmative response, even a possible experimental realization of Gödel's "contradictions" might be foreshadowed.[7] On the other hand, strong negative answers to the questions would en-

[6] Note Gödel's use of the strong word "contradiction," and his implied acknowledgment that "actual feasibility" of time-travel *would* constitute grounds for the rejection of such structures *a priori*. In my view, this concession is not warranted.

[7] I have remarked (n. 6 above) that "contradiction" is a strong word to use here. Recognition of the possibility of such temporal paradoxes in the framework of general relativity long antedates Gödel's discovery of definite models of this sort. The view that these paradoxes, even

This content downloaded from 147.026.011.080 on July 17, 2017 06:50:22 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

courage the view that the "paradoxical" structures may simply be ruled out, and would certainly weaken the force of Gödel's arguments.

In short, the demonstrated possibility of anomalies of temporal order in general relativity gives rise to the question: *to what extent the existence or non-existence of a univocal time-ordering along all time-like world-lines is susceptible of manipulation* (by the physical rearrangement of matter) *within a given cosmic model*; and a clear answer to this question would be of very great philosophic interest.

APPENDIX

Derivation of the Gödel Results

IV

The underlying space-time manifold of Gödel's model is topologically equivalent to Euclidean four-dimensional space: four differentiable coordinate-functions—z_0, z_1, z_2, z_3—are given on all of space-time. In terms of these coordinates, the metric ground-form of space-time is taken to be:[8]

$$(1) \qquad ds^2 = (dz_0 + e^{z_1}dz_2)^2 - \tfrac{1}{2}(dz_1^2 + e^{2z_1}dz_2^2 + dz_3^2);$$

This has the right signature for a Minkowski metric; the lines defined by constant values of z_1, z_2, z_3, are easily seen to be time-like geodesics (and proper time along them to be measured by z_0); and uniform flow along these "z_0-lines"—i.e. the one-parameter family of maps

$$\psi_t : (z_0, z_1, z_2, z_3) \to (z_0 + t, z_1, z_2, z_3)$$

—clearly leaves both the metric and the z_0-lines themselves invariant. The distribution of matter is taken to be uniform (i.e., we assume a constant space-time density μ for mass-time), and the world-lines of flow of matter are taken to be the z_0-lines;

---

when conceived as within the reach of direct experience, do *not* amount to contradictions, is expressed by Weyl ([8], p. 249): "From each world-point there proceeds the double cone of active future and passive past. Whereas in the special theory of relativity these are separated by an intermediate region, it is here [in the general theory] intrinsically quite possible that the cone of the active future encroaches upon that of the passive past; it can therefore happen, in principle, that I now experience events which arise, in part, only as effects of my future decisions and actions. Neither is it excluded that a world-line, though it has time-like direction at each point (in particular, the world-line of my body), returns to the neighborhood of a world-point that it has already once passed. From this there would result a more radical sort of Doppelgänger than ever an E. T. A. Hoffmann has imagined. In point of fact, such considerable variations of the g_{ik} as would be required for these effects do not occur in the world-region in which we live; yet to think through these possibilities, as they bear upon the philosophical problem of the relation of cosmic and phenomenal time, has a certain interest. However paradoxical what thus comes to light, a genuine contradiction to the facts given immediately in our experience nowhere arises." (Cf. also Reichenbach, [7], pp. 141–142.)

[8] To simplify later computations, I use coordinates slightly different from Gödel's; his are given in terms of mine by: $x_0 = \sqrt{2}z_0$; $x_1 = z_1$; $x_2 = \sqrt{2}z_2$; $x_3 = z_3$. I further take his constant a^2 to be $\frac{1}{2}$ (which amounts to a choice of unit for ds).

This content downloaded from 147.026.011.080 on July 17, 2017 06:50:22 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

in view of the foregoing, these two assumptions are compatible (the flow defined by the z_0-lines will not disturb a uniform density).

The curvature tensor of this metric is easily computed (details are given in [4]), and one finds that the assumptions made—metric defined by (1), uniform mass-time density, and matter-flow represented by the z_0-lines—are compatible with Einstein's gravitational equations (provided that both the cosmological constant and the mass-time density have the right value).[9]

V

To establish the chief point about the time-structure of this "world," we introduce a new system of parameters—ρ, θ, τ, η—related to the coordinate-functions z_i by the following conditions:[10]

$$(2) \qquad e^{z_1} = e^{\rho} \cos^2 \theta + e^{-\rho} \sin^2 \theta$$

$$(3) \qquad z_2 e^{z_1} = (e^{\rho} - e^{-\rho}) \sin \theta \cos \theta$$

$$(4) \qquad \tan\left(\theta + \frac{z_0 - \tau}{2}\right) = e^{-\rho} \tan \theta$$

and $(4')\ |z_0 - \tau| < \pi$

$$(5) \qquad z_3 = \eta.$$

It is important to make quite explicit what these relations are intended to signify. Note first that for every pair of values of ρ and θ, (2) and (3) determine unique values for z_1 and z_2; and that the dependence of z_1 and z_2 on ρ and θ is a continuous and differentiable one. (This follows easily from the fact that the right-hand side of (2) is always >0.) If, in addition, τ is specified, (4) tells us that $\theta + (z_0 - \tau)/2$ is a number with given tangent; and (4') says that this number differs from θ by less than $\pi/2$. Now, for the real numbers in any interval of length π (as arguments), the tangent assumes every real value and the value "∞" (counting $+\infty$ and $-\infty$ as the same) exactly once, except for one value which is assumed at both end-points of the interval. But the value of $\tan(\theta + (z_0 - \tau)/2)$ according to (4), namely $e^{-\rho} \tan \theta$, is certainly not assumed at the end-points of the interval $(\theta - \pi/2, \theta + \pi/2)$, since $\tan(\theta \pm \pi/2) = -1/\tan \theta$;[11] it is therefore assumed at a unique interior point of

[9] If the unit of mass is so chosen that the equations can be written:

$$R_{ij} - \tfrac{1}{2} g_{ij} R = T_{ij} + \lambda g_{ij};$$
$$T_{ij} = \mu u_i u_j;$$

(where the u_i are the covariant components of the unit tangent-vector of material flow), then (with the unit for ds already adopted) we must have $\lambda = -1$, $\mu = 2$.

[10] Again, for convenience in computation, I deviate from Gödel. His r, φ, t, y, are related to my ρ, θ, τ, η, by: $r = \rho/2$; $\varphi = 2\theta$; $t = \tau/\sqrt{2}$; $y = \eta/2$.

[11] Yet more explicitly: if $\tan \theta \neq 0$ and $\neq \infty$, then $e^{-\rho} \tan \theta$ has the same sign as $\tan \theta$, whereas $\tan(\theta \pm \pi/2)$—i.e. $-1/\tan \theta$—has the opposite sign; if $\tan \theta = 0$, then $e^{-\rho} \tan \theta = 0$, whereas $\tan(\theta \pm \pi/2) = \infty$; if $\tan \theta = \infty$, then $e^{-\rho} \tan \theta = \infty$, whereas $\tan(\theta \pm \pi/2) = 0$.

This content downloaded from 147.026.011.080 on July 17, 2017 06:50:22 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

that interval; thus (4) and (4′) are uniquely solvable for $\theta + (z_0 - \tau)/2$ as a function of ρ and θ. That the dependence is continuous and differentiable follows easily (by the implicit function theorem); and we see that z_0 is determined as a continuous and differentiable function of ρ, θ, τ. Equation (5) of course gives rise to no difficulty. We have, in summary, established the following proposition:

> Conditions (2) through (5), with (4′), determine the coordinates z_i as functions of the parameters ρ, θ, τ, η, defined for all real values of the latter, and continuous and differentiable everywhere.

It is evident that a change of θ by (addition of) π, with no changes in the other variables, will leave all the conditions undisturbed. We may therefore regard θ as ranging, not over the real numbers, but over the reals modulo π (the notions of continuity and differentiability remain applicable; in particular, the "differential form" $d\theta$ remains well-defined). Accordingly, we may and shall henceforth take ρ, θ, τ, η, as coordinate-functions on a "hypercylinder," the product of three copies of the real line (with coordinates ρ, τ, η) and a circle (with coordinate θ, scaled so that π represents a full circuit). We are thus led to read (2) through (5), with (4′), as defining *a continuous and differentiable mapping f of this hypercylinder into space-time.*[12]

It should be noted that the image, under the mapping f, of any "τ-line" of the hypercylinder—that is, a line characterized by constant values of ρ, θ, η—is a z_0-line (for constancy of ρ, θ, η, implies constancy of z_1, z_2, z_3). Along such a line, according to (4) and (4′), the difference $z_0 - \tau$ will be constant. In short, as we may (somewhat loosely) say: the τ-lines are z_0-lines; and along them τ, like z_0, measures proper time.

VI

Under f, any tangent-vector to the hypercylinder is mapped to a tangent-vector to space-time; ds^2 can therefore be evaluated for such a vector. We wish to determine this value in terms of the differential forms $d\rho$, $d\theta$, $d\tau$, $d\eta$ (which constitute a basis for all differential forms on the hypercylinder). The computation consists essentially in differentiating (2), (3), (4), and (5); but the details can be onerous. Gödel, in [4], does not present such a computation (which he says is "cumbersome"), indicating instead a somewhat sophisticated roundabout way to the result. It seems worth while to show here how a straightforward computation can be effected with little pain.

[12] The points at which $\rho = 0$ are the critical points of this mapping. Everywhere else, the mapping is locally invertible; and if we impose the further condition that $\rho > 0$, the inverse is unique. We then have ρ (≥ 0) defined and continuous on all of space-time, and differentiable except where $z_1 = z_2 = 0$; θ and τ defined, continuous, and differentiable, except where $z_1 = z_2 = 0$ (the function θ having as range the reals modulo π); and, of course, η defined, continuous, and differentiable everywhere. On the region characterized by $\rho \neq 0$, these four functions constitute a coordinate system; and, in this sense, (2) through (5) represent a "transformation to a new system of coordinate-functions." But for our purposes, none of this is needed: direct consideration of the mapping f will suffice.

This content downloaded from 147.026.011.080 on July 17, 2017 06:50:22 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

Differentiation of (2) gives:

$$e^{z_1}dz_1 = (e^\rho \cos^2\theta - e^{-\rho}\sin^2\theta)\,d\rho - 2(e^\rho - e^{-\rho})\sin\theta\cos\theta\,d\theta;$$

i.e.,

$$(6)\qquad dz_1 = \frac{e^\rho\cos^2\theta - e^{-\rho}\sin^2\theta}{e^{z_1}}\,d\rho - 2\frac{e^\rho - e^{-\rho}}{e^{z_1}}\sin\theta\cos\theta\,d\theta.$$

(Of course, using (2), this determines dz_1 explicitly in terms of ρ, θ, $d\rho$, $d\theta$; the denominators are left in terms of z_1 only for abbreviation.)

Differentiation of (3) gives:

$$(7)\qquad z_2e^{z_1}dz_1 + e^{z_1}dz_2 = (e^\rho + e^{-\rho})\sin\theta\cos\theta\,d\rho + (e^\rho - e^{-\rho})(\cos^2\theta - \sin^2\theta)\,d\theta.$$

From (3) and (6) together we have:

$$(8)\qquad z_2e^{z_1}dz_1 = \frac{(e^\rho - e^{-\rho})(e^\rho\cos^2\theta - e^{-\rho}\sin^2\theta)}{e^{z_1}}\sin\theta\cos\theta\,d\rho - 2\frac{(e^\rho - e^{-\rho})^2}{e^{z_1}}\sin^2\theta\cos^2\theta\,d\theta.$$

From (7) and (8) we obtain $e^{z_1}\,dz_2$; but simplifications occur. First, considering the coefficient of $\sin\theta\cos\theta\,d\rho$, we observe that this will be:

$$(e^\rho + e^{-\rho}) - \frac{(e^\rho - e^{-\rho})(e^\rho\cos^2\theta - e^{-\rho}\sin^2\theta)}{e^{z_1}};$$

i.e.

$$\frac{(e^\rho + e^{-\rho})(e^\rho\cos^2\theta + e^{-\rho}\sin^2\theta) - (e^\rho - e^{-\rho})(e^\rho\cos^2\theta - e^{-\rho}\sin^2\theta)}{e^{z_1}};$$

and this reduces to $2(\sin^2\theta + \cos^2\theta)/e^{z_1}$; i.e. to $2/e^{z_1}$. As for the coefficient of $(e^\rho - e^{-\rho})\,d\theta$, it will be:

$$\cos^2\theta - \sin^2\theta + 2\frac{(e^\rho - e^{-\rho})}{e^{z_1}}\sin^2\theta\cos^2\theta;$$

i.e.

$$\frac{(\cos^2\theta - \sin^2\theta)(e^\rho\cos^2\theta + e^{-\rho}\sin^2\theta) + 2(e^\rho - e^{-\rho})\sin^2\theta\cos^2\theta}{e^{z_1}}.$$

The numerator reduces to:

$$e^\rho\cos^2\theta\,(\cos^2\theta + \sin^2\theta) - e^{-\rho}\sin^2\theta\,(\cos^2\theta + \sin^2\theta);$$

i.e. $e^\rho\cos^2\theta - e^{-\rho}\sin^2\theta.$

Putting these results together, we find:

$$(9)\qquad e^{z_1}dz_2 = \frac{2\sin\theta\cos\theta}{e^{z_1}}\,d\rho + \frac{(e^\rho\cos^2\theta - e^{-\rho}\sin^2\theta)(e^\rho - e^{-\rho})}{e^{z_1}}\,d\theta.$$

This content downloaded from 147.026.011.080 on July 17, 2017 06:50:22 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

Now, the expression for ds^2 contains dz_1 and dz_2 in the combination: $dz_1^2 + e^{2z_1}\,dz_2^2$. Squaring (6) and (9), we see directly that the terms in $d\rho\,d\theta$ are equal and opposite; the sum contains only terms in $d\rho^2$ and $d\theta^2$. Both have e^{2z_1} as denominator. The numerator of the coefficient of $d\rho^2$ is:

$$(e^{2\rho}\cos^4\theta - 2\cos^2\theta\sin^2\theta + e^{-2\rho}\sin^4\theta) + 4\sin^2\theta\cos^2\theta;$$

i.e. $$(e^{\rho}\cos^2\theta + e^{-\rho}\sin^2\theta)^2;$$

i.e. $$e^{2z_1};$$

so the coefficient of $d\rho^2$ is unity. The numerator of the coefficient of $d\theta^2$ is:

$$(e^{\rho} - e^{-\rho})^2\,[4\sin^2\theta\cos^2\theta + (e^{2\rho}\cos^4\theta - 2\cos^2\theta\sin^2\theta + e^{-2\rho}\sin^4\theta)];$$

i.e. $$(e^{\rho} - e^{-\rho})^2 e^{2z_1};$$

so the coefficient of $d\theta^2$ is $(e^{\rho} - e^{-\rho})^2$. In sum, we have found:

(10) $$dz_1^2 + e^{2z_1}\,dz_2^2 = d\rho^2 + (e^{\rho} - e^{-\rho})^2\,d\theta^2.$$

Next, differentiating (4), we obtain:

$$\sec^2\left(\theta + \frac{z_0 - \tau}{2}\right)\left(d\theta + \frac{dz_0}{2} - \frac{d\tau}{2}\right) = -e^{-\rho}\tan\theta\,d\rho + e^{-\rho}\sec^2\theta\,d\theta.$$

But $$\sec^2\left(\theta + \frac{z_0 - \tau}{2}\right) = 1 + \tan^2\left(\theta + \frac{z_0 - \tau}{2}\right);$$

and this, by (4) itself, is equal to $1 + e^{-2\rho}\tan^2\theta$; so we have:

$$d\theta + \frac{dz_0}{2} - \frac{d\tau}{2} = -\frac{e^{-\rho}\tan\theta\,d\rho}{1 + e^{-2\rho}\tan^2\theta} + \frac{e^{-\rho}\sec^2\theta\,d\theta}{1 + e^{-2\rho}\tan^2\theta};$$

or (multiplying both numerators and both denominators on the right by $e^{\rho}\cos^2\theta$):

$$d\theta + \frac{dz_0}{2} - \frac{d\tau}{2} = -\frac{\sin\theta\cos\theta}{e^{\rho}\cos^2\theta + e^{-\rho}\sin^2\theta}\,d\rho + \frac{1}{e^{\rho}\cos^2\theta + e^{-\rho}\sin^2\theta}\,d\theta;$$

i.e.

(11) $$dz_0 = -2\frac{\sin\theta\cos\theta}{e^{z_1}}\,d\rho + 2\left(\frac{1}{e^{z_1}} - 1\right)d\theta + d\tau.$$

What occurs in ds^2 is the combination $dz_0 + e^{z_1}dz_2$. From (11) and (9) we see that the coefficient of $d\rho$ in this sum vanishes. For the coefficient of $d\theta$ we have:

$$\frac{(e^{\rho}\cos^2\theta - e^{-\rho}\sin^2\theta)(e^{\rho} - e^{-\rho}) + 2}{e^{z_1}} - 2.$$

But the numerator of the fraction is:

$$e^{2\rho}\cos^2\theta - \sin^2\theta - \cos^2\theta + e^{-2\rho}\sin^2\theta + 2;$$

This content downloaded from 147.026.011.080 on July 17, 2017 06:50:22 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

and this reduces to $e^{2\rho}\cos^2\theta + e^{-2\rho}\sin^2\theta + 1$; which is:

$$(e^{\rho}\cos^2\theta + e^{-\rho}\sin^2\theta)(e^{\rho} + e^{-\rho});$$

i.e. $$e^{z_1}(e^{\rho} + e^{-\rho}).$$

Thus the coefficient of $d\theta$ is just $e^{\rho} + e^{-\rho} - 2$, or $(e^{\rho/2} - e^{-\rho/2})^2$; and we have:

$$dz_0 + e^{z_1}\,dz_2 = (e^{\rho/2} - e^{-\rho/2})^2\,d\theta + d\tau. \tag{12}$$

Finally, then, (1), (5), (10), and (12) yield:

$$ds^2 = d\tau^2 - \tfrac{1}{2}\,d\rho^2 + [(e^{\rho/2} - e^{-\rho/2})^4 - \tfrac{1}{2}(e^{\rho} - e^{-\rho})^2]\,d\theta^2 - \tfrac{1}{2}\,d\eta^2 + 2(e^{\rho/2} - e^{-\rho/2})^2\,d\tau\,d\theta. \tag{13}$$

VII

The coefficient of $d\theta^2$ in the expression (13) for ds^2 is positive for ρ sufficiently large—namely, for $e^{\rho/2} - e^{-\rho/2} > 2$. (Indeed, this coefficient is equal to:

$$\begin{aligned}&(e^{\rho/2} - e^{-\rho/2})^4 - \tfrac{1}{2}[(e^{\rho/2} + e^{-\rho/2})(e^{\rho/2} - e^{-\rho/2})]^2\\ &= (e^{\rho/2} - e^{-\rho/2})^2[(e^{\rho/2} - e^{-\rho/2})^2 - \tfrac{1}{2}(e^{\rho/2} + e^{-\rho/2})^2]\\ &= \tfrac{1}{2}(e^{\rho/2} - e^{-\rho/2})^2[(e^{\rho/2} - e^{-\rho/2})^2 - 4].)\end{aligned}$$

Consider, then, a mapping p of the unit interval [0, 1] into the hypercylinder—a "parametrized path"—taking the real number x to the point $p(x)$ characterized by:

$$\rho(p(x)) = R; \qquad \tau(p(x)) = T;$$
$$\theta(p(x)) = \pi x; \qquad \eta(p(x)) = Y;$$

where R, T, and Y, are constants, with $e^{R/2} - e^{-R/2} > 2$. This is clearly a closed path; that is, $p(0) = p(1)$; and its image under f is therefore a closed path, fp, in space-time. Along p we have $d\tau = d\rho = d\eta = 0$, $d\theta = \pi\,dx$; hence we have along fp:

$$ds^2 = [(e^{R/2} - e^{-R/2})^4 - \tfrac{1}{2}(e^{R} - e^{-R})^2]\pi^2\,dx^2;$$

and since this is everywhere positive, *the path fp is everywhere time-like.* Furthermore, we have for space-time a globally coherent separation of all time-like non-zero vectors into "future-pointing" and "past-pointing" ones: at any point, that lobe of the light-zone which contains the unit tangent-vector to the line of material flow (the z_0-line in the direction of increasing z_0) is the "future" lobe. From the fact that the coefficient of $d\tau\,d\theta$ in (13) is positive, it then follows that *the tangent-vector to fp* (which is a "θ-line") *in the direction of increasing x* (increasing θ) *is always future-pointing.* Thus fp is a world-line representing a possible "history," or space-time course of a body; and it has the property that after the lapse of proper time amounting to:

$$[(e^{R/2} - e^{-R/2})^4 - \tfrac{1}{2}(e^{R} - e^{-R})^2]^{1/2}\pi,$$

This content downloaded from 147.026.011.080 on July 17, 2017 06:50:22 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

the body occupies the *same* point of space-time as at the start.[13] If we make the obvious extension of this parametrized path to all real parameter-values, the history of a body with this path as world-line will be *cyclic*—despite the fact that the histories of the "cosmic" masses, represented by the z_0-lines, are not cyclic. Moreover, it is obvious that a mild perturbation of the path p can change fp so that, although it remains time-like and future-pointing, its end-point is on the same z_0-line as, and (as measured by z_0) *earlier than*, its beginning: it suffices to replace the condition $\tau(p(x)) = T$ by $\tau(p(x)) = T - ax$, with a a positive constant smaller than $\pi[(e^{R/2} - e^{-R/2})^2 - 1/\sqrt{2}(e^R - e^{-R})]$ (that this is indeed positive is clear, since we know that $(e^{R/2} - e^{-R/2})^4 > \frac{1}{2}(e^R - e^{-R})^2$). Extending this path (in the obvious way) to all real parameter-values, we have the extraordinary picture of *a body whose history, winding about space-time so as always to be oriented towards the local future, encounters periodically*—as the body "ages"—*the same cosmic masses, at successively earlier stages of their own histories.* Finally, it is evident that a path of this "backwards-winding" sort can be pieced together (smoothly) with one that follows the general trend of a z_0-line, in quite arbitrary fashion; obtaining a possible history which "visits its own past" *ad libitum.*

[13] We have assumed that the z_0-lines represent the "histories" of matter; but this is meant only for the average "cosmic" situation. The introduction of small bodies moving in other ways will have a negligible effect upon the metric of space-time.

REFERENCES

[1] Chandrasekhar, S., and Wright, J. P., "The Geodesics in Gödel's Universe," *Proceedings of the National Academy of Sciences of the United States of America*, vol. 47, No. 3, 1961 (March), pp. 341–347.

[2] Earman, J., "On Going Backward in Time," *Philosophy of Science*, vol. 34, no. 3, 1967 (September), pp. 211–222.

[3] Einstein, A., "Remarks Concerning the Essays Brought Together in this Co-operative Volume," *Albert Einstein: Philosopher-Scientist* (ed. Paul Arthur Schilpp), The Library of Living Philosophers, Inc., Evanston, Illinois, 1949, pp. 665–688.

[4] Gödel, K., "An Example of a New Type of Cosmological Solutions of Einstein's Field Equations of Gravitation," *Reviews of Modern Physics*, vol. 21, no. 3, 1949 (July), pp. 447–450.

[5] Gödel, K., "A Remark about the Relationship between Relativity Theory and Idealistic Philosophy," *Albert Einstein: Philosopher-Scientist* (ed. Paul Arthur Schilpp), The Library of Living Philosophers, Inc., Evanston, Illinois, 1949, pp. 557–562.

[6] Grünbaum, A., *Philosophical Problems of Space and Time*, Alfred A. Knopf, New York, 1963.

[7] Reichenbach, H., *The Philosophy of Space and Time* (tr. by Maria Reichenbach and John Freund), Dover Publications, Inc., New York, 1957.

[8] Weyl, H., *Raum Zeit Materie*, 4th edit., Verlag von Julius Springer, Berlin, 1921.

This content downloaded from 147.026.011.080 on July 17, 2017 06:50:22 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).